Shiza Yasar

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EDUCATION

University of La Verne: Bachelor's in computer science, minor: software, and engineering. | GPA: 3.85 **Expected May '29**

PROFESSIONAL WORK EXPERIENCE

Code Ninjas: *Code Sensei* **Jan '25 – Jun '25**

- Delivered hands-on coding lessons for students ages 5–12 using Python, Unity 3D, Roblox Studio, Minecraft Education, MakeCode Arcade, and block-based programming, helping students progress from basic concepts to fully playable games.
- Independently led a 6-week Minecraft Education program across three schools, using game-based exercises and guided debugging to improve student completion speed for introductory coding activities by an estimated 25%.
- Co-led a FIRST LEGO League (FLL) team, supporting students with research, robot programming, testing, and collaborative problem solving.

Abstron: *Game Development Intern* **Aug '24 – Jan '25**

- Authored a 16-page Game Design Document for a virtual reality cooking simulator, defining game mechanics, levels, player interactions, and development requirements to give the production team a clear implementation roadmap.
- Designed user-interface concepts for multiple game modes and settings, focusing on clear navigation, accessibility, and player usability.
- Sourced game assets, animations, and sound effects to support a cohesive and immersive player experience

CYTE Foundation: *STEM Instructor* **Jun '21 – Jun '24**

- Taught introductory robotics to 50+ students through hands-on building, programming, and troubleshooting exercises, helping students move from foundational concepts to functioning robot projects.
- Guided 50+ students through iterative programming, testing, and debugging, resulting in every student completing a functional robotics build.
- Helped secure funding and establish fully equipped computer labs, expanding access to computers, robotics tools, and hands-on technology learning across partner schools.

Inno Social: *ICT Head* **Jul '22 – Jul '23**

- Developed and delivered practical ICT curriculum across multiple schools and orphanages, teaching Microsoft Word, PowerPoint, Excel, and digital literacy through structured, task-based lessons.
- Improved student task-completion speed for common ICT assignments by an estimated 20% through step-by-step instruction, demonstrations, and guided practice.

PROJECTS

3D-Printed Robotic Arm with Computer Vision Control **Jul '26 – Sep '26**

- Developed and assembled a multi-part robotic arm using Bambu Studio, Bambu Lab A1, servo motors, and 3D-printed components for human-motion replication.
- Integrated Python computer vision using OpenCV and MediaPipe hand tracking to translate hand and finger movements into real-time servo motion through an Arduino (pyFirmata2).
- Optimized wall count, infill, supports, and print orientation to improve structural reliability while reducing failed prints and unnecessary material usage.
- Troubleshot servo calibration, mechanical alignment, filament flow, and component fit to improve movement consistency and reduce repeated assembly adjustments by 30%.

Dusk Catchers: Co-op Browser Game | C++ **Oct '26 – Present**

- Developing a cozy two-player co-op browser game in C++17 with raylib, where players share a glowing string to catch fireflies and carry them home to light village lanterns.
- Designing a carry mechanic that tracks string speed and stretch each frame, shaking fireflies loose when players move too fast, with line-vs-circle collision for catching.
- Structuring gameplay with object-oriented C++ classes and data-driven levels, so each of seven evenings loads from a data file instead of requiring new code.
- Building local co-op and solo modes for one keyboard or gamepads, using CMake for builds and Emscripten to publish a playable web version.

SKILLS

Languages: C++, C#, Python, SQL, JavaScript, HTML, CSS
Software: Unity 3D, Visual Studio Code, Bambu Studio, Fusion 360
Development: OpenCV, MediaPipe, NumPy, React, Vite, Roblox Studio, MakeCode Arcade
Hardware / Robotics: Arduino, Servo Motors, Gyroscope Sensors, 3D Printing
Other: UI Design, Game Design, Object-Oriented Programming, Debugging